

===== Bank Client
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pki@pky:~/bank\$ = openssl genpkey -algorithm RSA -out bank_pri.pem

pki@pky:~/bank\$ = openssl rsa -pubout -in bank_pri.pem -out bank_pub.pem

pki@pky:~/bank/acct1\$ = openssl genpkey -algorithm RSA -out acct1_pri.pem

pki@pky:~/bank/acct1\$ = openssl rsa -pubout -in acct1_pri.pem -out acct1_pub.pem

pki@pky:~/bank/acct1\$ = openssl pkeyutl -encrypt -pubin -inkey ../bank_pub.pem -in msg.txt -out msg.enc

mv msg.enc ../

pki@pky:~/bank\$ = openssl pkeyutl -decrypt -inkey bank_pri.pem -in msg.enc -out msg.txt

pki@pky:~/bank\$ = ls

acct1 acct2 acct3 bank_pri.pem bank_pub.pem msg.enc msg.txt

===== RSA Sign and Verify

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pki@pky:~/bank\$ = openssl genpkey -algorithm RSA -out pri.pem

pki@pky:~/bank\$ = openssl rsa -pubout -in bank_pri.pem -out pub.pem

. openssl dgst -sha256 -sign keyfile.pem -out file1.txt.sha256.sig file1.txt

. openssl dgst -sha256 -verify public_key.pem -signature file1.txt.sha256.sig
file1.txt

===== Create certificates rootca,sub,private
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Root CA Certificate

· Generate root CA private key:

·= openssl genpkey -algorithm RSA -out root.key

· Generate root CA Certificate Signing Request (CSR):

·= openssl req -new -key root.key -out root.csr

· Self-sign the Root CA certificate:

= openssl x509 -req -in root.csr -signkey root.key -out root.crt -days 3650

· View the file in a readable format:

·= openssl x509 -in root.crt -text -noout

Sub CA Certificate

· Generate sub CA private key:

·= openssl genpkey -algorithm RSA -out sub.key

· Generate sub CA Certificate Signing Request (CSR):

·= openssl req -new -key sub.key -out sub.csr

· Sign the sub CA CSR with Root CA:

·= openssl x509 -req -in sub.csr -CA root.crt -CAkey root.key -CAcreateserial -out sub.crt -days 1825

· View the file in a readable format:

·= openssl x509 -in sub.crt -text -noout

Issue Digital Certificates from Sub CA

· Generate some server's private key:

·= openssl genpkey -algorithm RSA -out server.key

· Generate CSR for an end entity (e.g. some server): cdac.ac.in

·= openssl req -new -key server.key -out server.csr

· Sign the CSR with sub CA:

·= openssl x509 -req -in server.csr -CA sub.crt -CAkey sub.key -CAcreateserial -out server.crt -days 365

· View the file in a readable format:

·= openssl x509 -in server.crt -text -noout